

CHEROKEE, IA, USA

WMO: 720344

Lat: 42.732N

Lon: 95.556W

Elev: 374

StdP: 96.92

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 54920

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.4	-19.5	-26.2	0.4	-21.5	-23.6	0.5	-19.1	11.4	-4.5	10.6	-4.2	1.4	0	0.541

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	33.3	23.2	32.0	22.7	30.2	21.9	25.7	30.2	24.7	29.2	23.8	28.2	5.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.6	20.5	28.5	23.2	18.9	27.3	22.5	18.0	26.5	81.7	29.7	77.0	29.4	73.5	28.1	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.5	9.0	8.1	-27.9	35.8	3.6	2.0	-30.5	37.2	-32.6	38.4	-34.6	39.6	-37.3	41.0		
			-28.5	27.6	3.4	0.7	-31.0	28.1	-32.9	28.4	-34.8	28.8	-37.3	29.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.2	-5.9	-5.8	2.9	8.9	15.6	21.6	23.6	21.8	17.8	10.1	2.8	-4.3
	DBStd	11.92	7.23	6.73	7.20	5.73	5.13	3.47	3.42	3.13	4.45	5.25	5.76	6.25
	HDD10.0	2004	493	443	239	87	11	0	0	0	2	63	224	442
	HDD18.3	3820	752	676	478	287	116	11	3	7	63	260	467	701
	CDD10.0	1702	0	0	21	53	184	348	422	366	235	67	7	0
	CDD18.3	476	0	0	1	3	31	109	166	114	46	6	0	0
	CDH23.3	4693	0	0	14	67	351	1081	1711	983	422	61	2	0
	CDH26.7	1634	0	0	2	21	116	369	700	299	112	14	0	0
Wind	WSAvg	3.4	3.6	3.5	3.9	4.3	3.8	3.1	2.6	2.4	2.8	3.4	3.5	3.4
Precipitation	PrecAvg	766	20	21	46	80	108	130	91	99	81	46	38	20
	PrecMax	925	39	47	101	145	181	218	192	275	195	117	110	44
	PrecMin	578	4	4	0	21	36	25	19	6	25	9	2	1
	PrecStd	104	9	11	31	32	38	53	38	71	41	30	27	12
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.3	14.7	25.5	30.3	32.9	34.2	36.8	34.3	32.2	28.8	21.4	12.4
		MCWB	6.0	9.6	14.4	16.2	19.6	23.0	23.6	23.7	21.3	19.3	13.3	7.2
	2%	DB	7.2	8.6	21.0	24.8	29.9	32.1	34.1	31.8	29.8	24.9	17.5	9.2
		MCWB	3.9	5.6	13.2	14.4	18.5	21.5	23.7	23.8	21.8	16.2	10.9	6.4
	5%	DB	4.8	5.8	17.1	21.6	27.4	30.6	32.4	29.9	27.7	22.0	14.4	6.1
		MCWB	2.5	3.3	12.5	13.2	17.8	20.9	24.0	23.2	20.1	15.4	9.0	3.5
	10%	DB	2.8	2.9	13.2	18.8	24.9	28.9	30.7	27.9	25.9	19.1	11.4	3.6
		MCWB	1.3	1.3	9.8	11.3	16.9	20.5	23.3	21.9	19.4	13.4	7.5	2.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.4	10.5	17.2	17.7	22.8	25.8	27.7	26.4	24.8	21.4	15.1	10.0
		MCDB	10.5	14.7	22.8	25.2	28.5	30.4	32.0	31.3	28.6	25.9	18.7	11.0
	2%	WB	4.1	5.7	15.1	15.8	21.1	24.2	26.3	25.2	23.6	18.3	12.1	5.7
		MCDB	6.5	8.7	18.1	22.0	26.5	28.8	30.6	29.5	27.1	23.0	17.1	8.2
	5%	WB	2.8	3.5	12.5	14.2	19.6	23.3	25.2	24.2	22.6	16.1	9.8	3.8
		MCDB	4.5	5.5	16.7	19.7	24.8	28.3	29.8	28.1	25.8	19.8	13.4	5.8
	10%	WB	1.5	1.6	9.7	12.6	18.1	22.1	24.1	23.2	21.3	13.9	7.7	2.4
		MCDB	2.7	3.0	13.9	17.8	22.3	26.8	29.0	26.8	24.1	17.8	11.4	3.9
Mean Daily Temperature Range	5% DB	MDBR	9.5	10.1	10.6	12.0	12.2	11.7	11.5	11.1	12.4	11.6	11.2	8.9
		MCDBR	11.8	13.1	16.5	17.2	15.4	13.4	12.8	12.4	13.6	15.4	15.6	12.2
		MCWBR	8.9	9.7	9.4	9.0	6.6	5.6	5.3	5.8	6.2	7.9	9.3	9.0
	5% WB	MCDBR	10.6	12.2	14.0	13.9	12.0	10.9	10.7	10.2	10.8	11.7	14.6	10.8
		MCWBR	8.3	9.2	8.4	8.1	6.5	5.9	5.6	5.8	6.1	7.5	9.4	8.5
Clear-Sky Solar Irradiance	taub	0.282	0.292	0.324	0.369	0.394	0.406	0.417	0.410	0.369	0.335	0.296	0.281	
	taud	2.404	2.394	2.398	2.299	2.300	2.325	2.280	2.291	2.373	2.450	2.484	2.439	
	Ebn at Noon	863	914	921	897	880	867	853	847	859	851	843	832	
	Edh at Noon	77	92	104	125	128	126	130	125	106	87	71	68	
All-Sky Solar Radiation	RadAvg	1.91	2.98	3.82	4.81	5.64	6.27	6.59	5.51	4.46	3.05	2.09	1.57	
	RadStd	0.18	0.23	0.33	0.42	0.41	0.49	0.33	0.39	0.28	0.38	0.13	0.12	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page